HARD · ALGEBRA · ANSWER KEY

Linear functions

01

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ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1 B

B. -4

Choice B is correct. It’s given that $f(3x) = x - 6$ for all values of x. If $3x = 6$, then $f(3x)$ will equal $f(6)$. Dividing both sides of $3x = 6$ by 3 gives $x = 2$. Therefore, substituting 2 for x in the given equation yields $f(3 \times 2) = 2 - 6$, which can be rewritten as $f(6) = -4$.

Choice A is incorrect. This is the value of the constant in the given equation for f. Choice C is incorrect and may result from substituting $x = 6$, rather than $x = 2$, into the given equation. Choice D is incorrect. This is the value of x that yields $f(6)$ for the left-hand side of the given equation; it’s not the value of $f(6)$.

Q2 .0465

Accept also: 2/43

The correct answer is $\frac{2}{43}$. It's given that $f(cx) = x - 8$ for all values of x, where c is a positive constant, and $f(2) = 35$. Therefore, for the given function f, $cx = 2$. Dividing both sides of this equation by c yields $x = \frac{2}{c}$. Substituting $\frac{2}{c}$ for x in the equation $f(cx) = x - 8$ yields $f\left(\frac{2c}{c}\right) = \frac{2}{c} - 8$, or $f\left(2\right) = \frac{2}{c} - 8$. Since it’s given that $f(2) = 35$, substituting 35 for $f(2)$ yields $35 = \frac{2}{c} - 8$. Adding 8 to both sides of this equation yields $43 = \frac{2}{c}$. Multiplying both sides of this equation by c yields $43c = 2$. Dividing both sides of this equation by 43 yields $c = \frac{2}{43}$. Note that 2/43, .0465, 0.046, and 0.047 are examples of ways to enter a correct answer.

Q3

515

The correct answer is 515. It's given that the equation $h = \frac{9(v - 273.15)}{5} + 32$ gives the corresponding temperature h, in degrees Fahrenheit, of any substance that has a temperature of v kelvins, where $v > 0$. Substituting 467.33 for h in the given equation yields $467.33 = \frac{9(v - 273.15)}{5} + 32$. Subtracting 32 from both sides of this equation yields $435.33 = \frac{9(v - 273.15)}{5}$. Multiplying both sides of this equation by 5 yields $2{,}176.65 = 9(v - 273.15)$. Dividing both sides of this equation by 9 yields $241.85 = v - 273.15$. Adding 273.15 to both sides of this equation yields $515 = v$. Therefore, if a substance has a temperature of 467.33 degrees Fahrenheit, the corresponding temperature, in kelvins, of this substance is 515.

Q4

C

C. $f(t) = -\frac{21}{130}t + 4$

Choice C is correct. It is assumed that the oil and gas production decreased at a constant rate. Therefore, the function f that best models the production t years after the year 2000 can be written as a linear function, $f(t) = mt + b$, where m is the rate of change of the oil and gas production and b is the oil and gas production, in millions of barrels, in the year 2000. Since there were 4 million barrels of oil and gas produced in 2000, $b = 4$. The rate of change, m, can be calculated as $\frac{4 - 1.9}{0 - 13} = -\frac{2.1}{13}$, which is equivalent to $-\frac{21}{130}$, the rate of change in choice C.

Choices A and B are incorrect because each of these functions has a positive rate of change. Since the oil and gas production decreased over time, the rate of change must be negative. Choice D is incorrect. This model may result from misinterpreting 1.9 million barrels as the amount by which the production decreased.

Q5

A

A. 16,250

Choice A is correct. Let the economist's model be the linear function $Q = mP + b$, where Q is the demand, P is the selling price, m is the slope of the line, and b is the y-coordinate of the y-intercept of the line in the xy-plane, where $y = Q$. Two pairs of the selling price P and the demand Q are given. Using the coordinate pairs (P, Q), two points that satisfy the function are $(40, 20{,}000)$ and $(60, 15{,}000)$. The slope m of the function can be found using the formula $m = \frac{Q_2 - Q_1}{P_2 - P_1}$. Substituting the given values into this formula yields $m = \frac{15{,}000 - 20{,}000}{60 - 40}$, or $m = -250$. Therefore, $Q = -250P + b$. The value of b can be found by substituting one of the points into the function. Substituting the values of P and Q from the point $(40, 20{,}000)$ yields $20{,}000 = -250(40) + b$, or $20{,}000 = -10{,}000 + b$. Adding 10,000 to both sides of this equation yields $b = 30{,}000$. Therefore, the linear function the economist used as the model is $Q = -250P + 30{,}000$. Substituting 55 for P yields $Q = -250(55) + 30{,}000 = 16{,}250$. It follows that when the selling price is $55 per unit, the demand is 16,250 units.

Choices B, C, and D are incorrect and may result from calculation or conceptual errors.

Q6

A

A. $f x = -d - cx$

Choice A is correct. It’s given that the graph of the linear function $y = f x + 19$ is shown. This means that the graph of $y = f x + 19$ can be translated down 19 units to create the graph of $y = f x$ and the y-coordinate of every point on the graph of $y = f x + 19$ can be decreased by 19 to find the resulting point on the graph of $y = f x$. The y-intercept of the graph of $y = f x + 19$ is $0, 3$. Translating the graph of $y = f x + 19$ down 19 units results in a y-intercept of the graph of $y = f x$ at the point $0, 3 - 19$, or $0, - 16$. The graph of $y = f x + 19$ slants down from left to right, so the slope of the graph is negative. The translation of a linear graph changes its position, but does not change its slope. It follows that the slope of the graph of $y = f x$ is also negative. The equation of a linear function f can be written in the form $f x = b + mx$, where b is the y-coordinate of the y-intercept and m is the slope of the graph of $y = f x$. It's given that c and d are positive constants. Since the y-coordinate of the y-intercept and the slope of the graph of $y = f x$ are both negative, it follows that $f x = - d - cx$ could define f.

Choice B is incorrect. This could define a linear function where its graph has a positive, not negative, y-intercept.

Choice C is incorrect. This could define a linear function where its graph has a positive, not negative, slope.

Choice D is incorrect. This could define a linear function where its graph has a positive, not negative, y-intercept and a positive, not negative, slope.

Q7

D

D. 192

Choice D is correct. The table gives that $f(x) = 0$ when $x = 2$. Substituting 0 for fx and 2 for x into the equation $f(x) = ax + b$ yields $0 = 2a + b$. Subtracting $2a$ from both sides of this equation yields $b = -2a$. The table gives that $f(x) = -64$ when $x = 1$. Substituting $-2a$ for b, -64 for fx, and 1 for x into the equation $f(x) = ax + b$ yields $-64 = a1 + -2a$. Combining like terms yields $-64 = -a$, or $a = 64$. Since $b = -2a$, substituting 64 for a into this equation gives $b = -264$, which yields $b = -128$. Thus, the value of $a - b$ can be written as $64 - (-128)$, which is 192.

Choice A is incorrect. This is the value of $a + b$, not $a - b$.

Choice B is incorrect. This is the value of $a - 2$, not $a - b$.

Choice C is incorrect. This is the value of $2a$, not $a - b$.

Q8

3331

The correct answer is 3,331. For engine speeds between 1,000 revolutions per minute (rpm) and 6,000 rpm, it's given that the equation $f\left(x\right) = \frac{1}{7}\left(x - a\right) + 433$ defines the linear function f and gives the predicted power, in brake horsepower (bhp), at an engine speed of x rpm, where a is a constant. It's also given that the car's predicted power is 433 bhp at an engine speed of 3,331 rpm. Substituting 3,331 for x and 433 for $f(x)$ in the equation yields $433 = \frac{1}{7}\left(3{,}331 - a\right) + 433$. Subtracting 433 from both sides of this equation yields $0 = \frac{1}{7}\left(3{,}331 - a\right)$. Multiplying both sides of this equation by 7 yields $0 = 3{,}331 - a$. Adding a to both sides of this equation yields $a = 3{,}331$. Thus, the value of a is 3,331.

Q9 A

A. $C(d) = 26d + 26$

Choice A is correct. It's given that the cost of renting a carpet cleaner is $\$52$ for the first day and $\$26$ for each additional day. Therefore, the cost Cd, in dollars, of renting the carpet cleaner for d days is the sum of the cost for the first day, $\$52$, and the cost for the additional $d - 1$ days, $\$26d - 1$. It follows that $Cd = 52 + 26d - 1$, which is equivalent to $Cd = 52 + 26d - 26$, or $Cd = 26d + 26$.

Choice B is incorrect. This function gives the cost of renting a carpet cleaner for d days if the cost is $\$78$, not $\$52$, for the first day and $\$26$ for each additional day.

Choice C is incorrect. This function gives the cost of renting a carpet cleaner for d days if the cost is $\$26$, not $\$52$, for the first day and $\$52$, not $\$26$, for each additional day.

Choice D is incorrect. This function gives the cost of renting a carpet cleaner for d days if the cost is $\$130$, not $\$52$, for the first day and $\$52$, not $\$26$, for each additional day.

Q10 A

A. 3.78

Choice A is correct. It's given that the function $Fx = \frac{9}{5}x - 273.15 + 32$ gives the temperature, in degrees Fahrenheit, that corresponds to a temperature of x kelvins. A temperature that increased by 2.10 kelvins means that the value of x increased by 2.10 kelvins. It follows that an increase in x by 2.10 increases $F(x)$ by $\frac{9}{5}2.10$, or 3.78. Therefore, if a temperature increased by 2.10 kelvins, the temperature increased by 3.78 degrees Fahrenheit.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

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